

Parameter	Symbol	Value
Vehicle bounding box		1.2 × 1.2 × 1.2 m
Clearance inflation margin	δ	0.4 m
Path safety extension		1.5 m
Gain-model sensor range	$r_{\max}$	8 m
Gain-model ray spacing		0.12 rad
Local sampling space		±15 m cube
Maximum local graph size		700 vertices
Maximum edge length		2.0 m
Executed path truncation		8 m
Gain weight (applied to the normalised gain $\hat{g}$)	w_g	60
Heading-change penalty weight	w_ψ	0.1
Path-length penalty weight	w_d	0.04
Cruise / homing speed limit		2.0 / 0.9 m s ⁻¹
Yaw rate limit		0.15 rad s ⁻¹
Exploration time budget		1,800 s
Onboard map voxel size	v	0.25 m
Map ray integration length	r_{map}	15 m
ESDF truncation distance		2.0 m